Industrial Internship Report on

" Banking Information System"

Prepared by

Avadhoot Tople

Dr. D.Y. Patil School of M.C.A

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Banking Information System". The system was developed to automate banking operations including account management, customer authentication, transactions, balance inquiry, fund transfer, reporting, and account status management. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

The six-week industrial internship organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT) provided an excellent opportunity to gain practical exposure to industry-oriented project development and implementation practices.

Internships play an important role in career development because they help students bridge the gap between academic learning and industrial expectations. They provide practical understanding of technologies, problem-solving approaches, project management practices, and real-world implementation methods.

The project assigned during this internship was “Banking Information System”. The project aimed to automate banking operations and manage customer accounts, transactions, authentication, account status, and reporting functionalities through a centralized software solution.

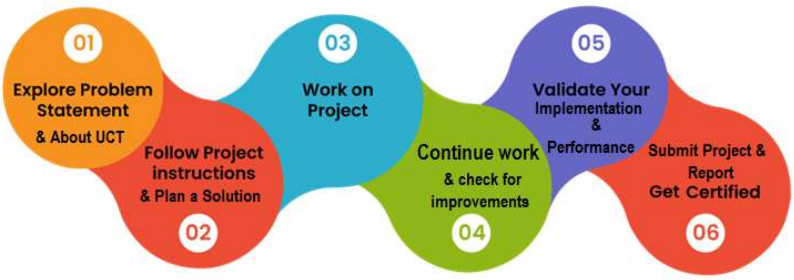
The internship program was systematically planned over six weeks where the initial phase involved understanding project requirements and system design, followed by implementation of different modules, integration of functionalities, testing procedures, validations, and documentation activities.

Throughout this internship, various concepts such as Core Java programming, object-oriented design, collections framework, file handling, validations, exception handling, authentication systems, and transaction processing were explored and implemented.

This internship improved technical knowledge, logical thinking ability, software design understanding, and practical implementation skills while also providing exposure to industrial workflows and development methodologies.

I would like to express my sincere gratitude to upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd (UCT), mentors, faculty members, and everyone who directly or indirectly supported throughout this internship journey.

To fellow students and juniors, internships provide valuable exposure and should be utilized as an opportunity to learn practical implementation skills and gain industry experience.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

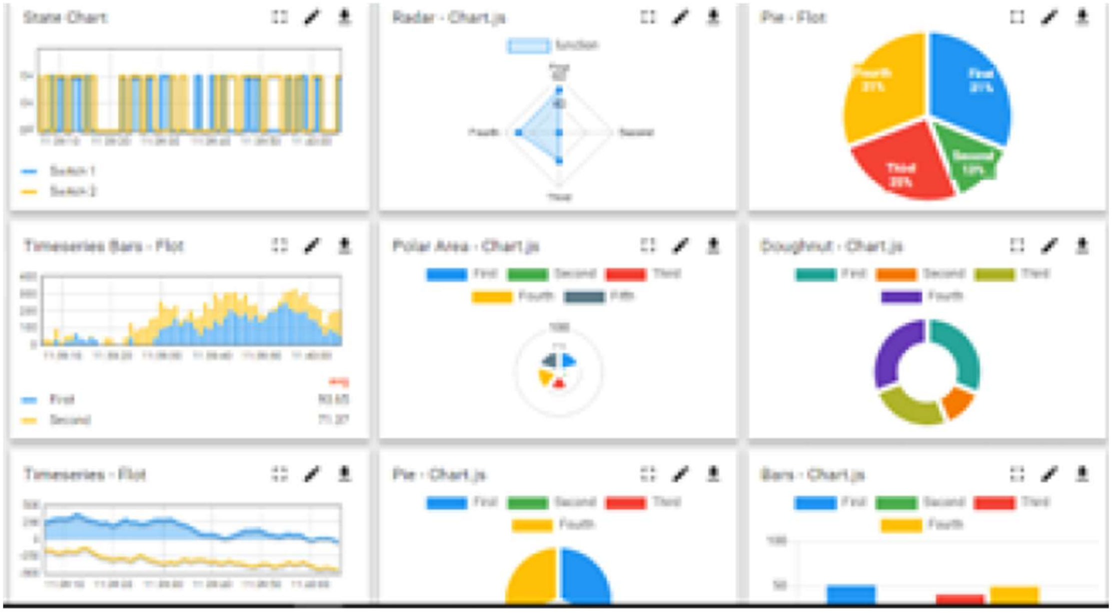
i. UCT IoT Platform ( )

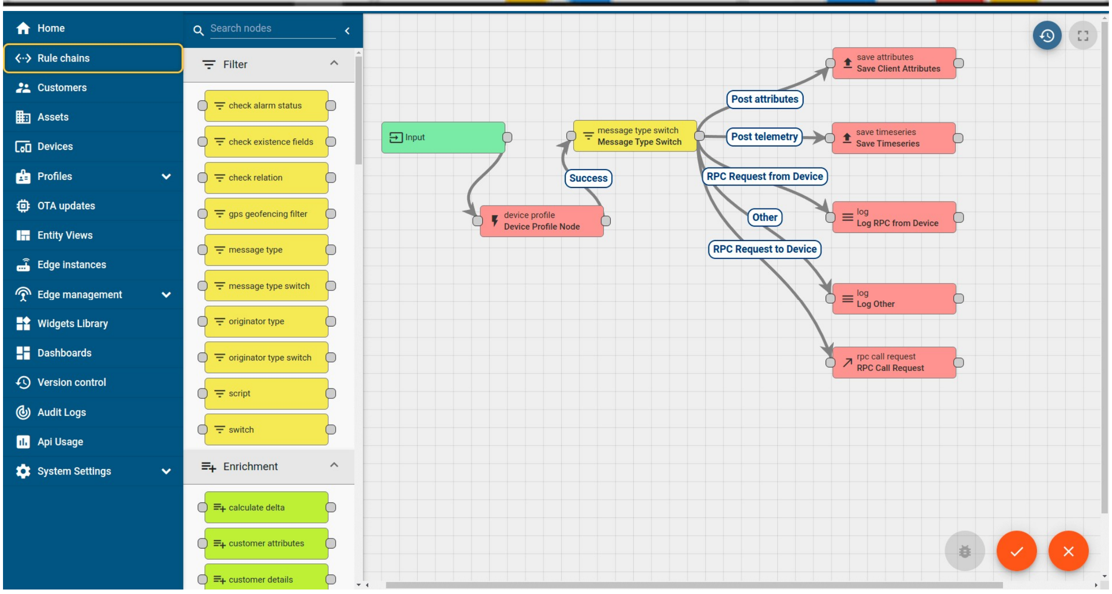
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

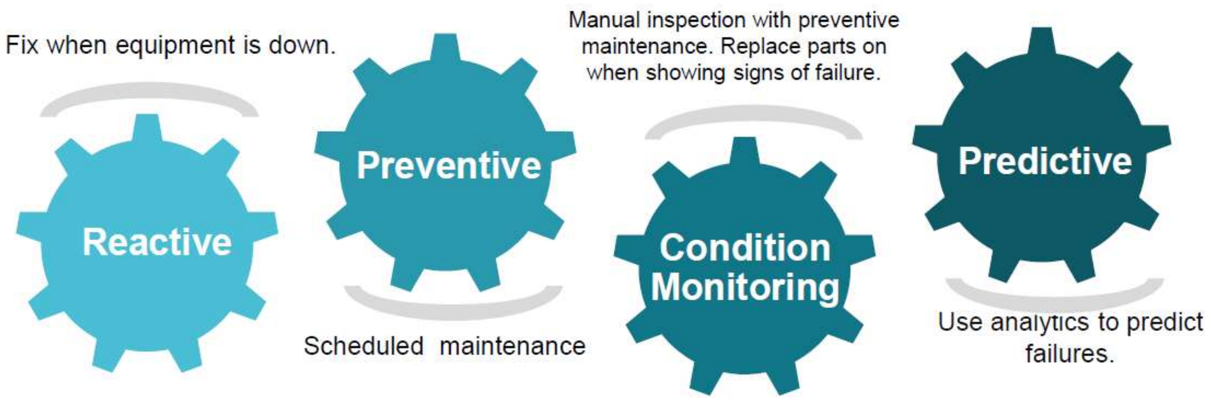
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

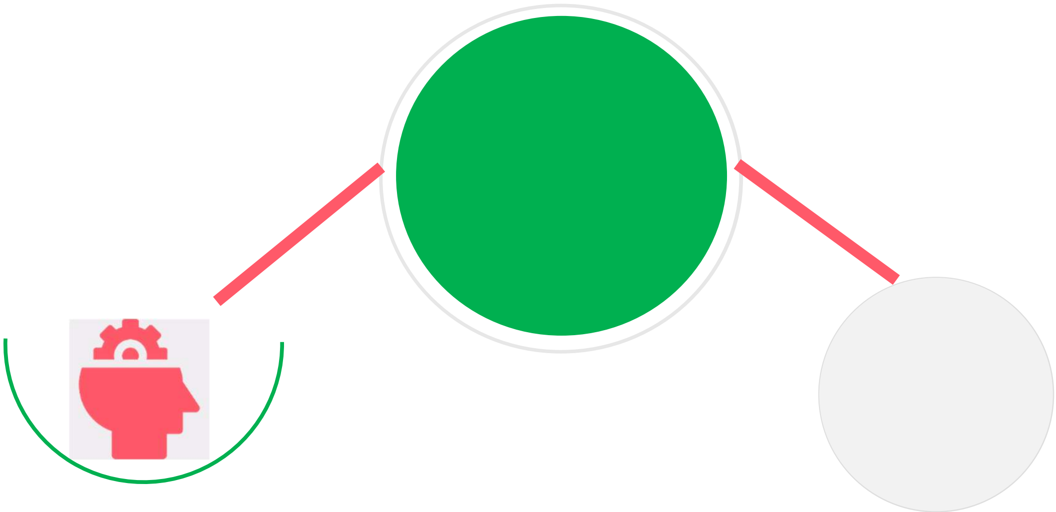
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

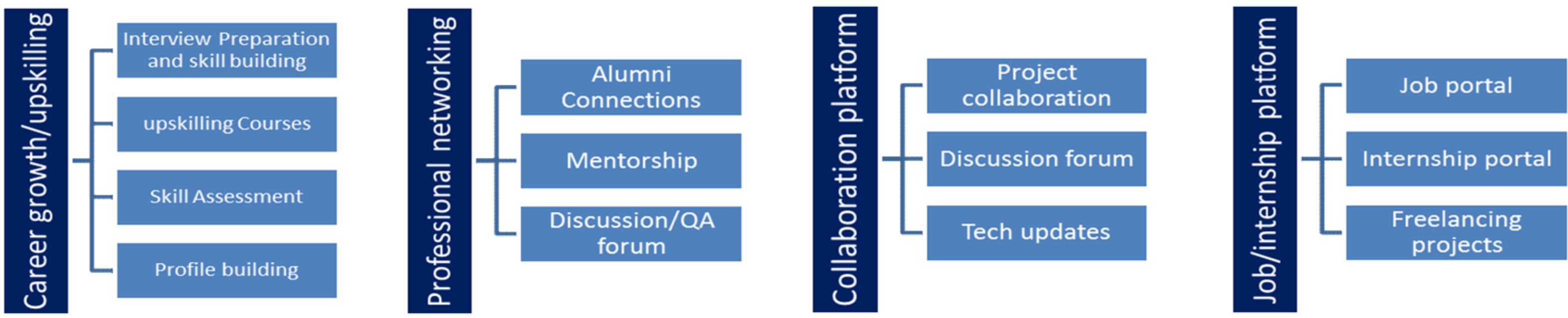
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] **GitHub Repository:**

https://github.com/Avadhoot-007/upskillcampus

[2] upskill Campus:

https://www.upskillcampus.com/

[3] Oracle Java Documentation:

https://docs.oracle.com/en/java/

2.6 Glossary

Terms	Acronym
BankingInformationSystem	BIS
UniConvergeTechnologies	UCT
upskill Campus	USC
Object Oriented Programming	OOP
Transaction ID	TXN
Application Programming Interface	API
Graphical User Interface	GUI
Java Development Kit	JDK
Integrated Development Environment	IDE
Java Database Connectivity	JDBC

3 Problem Statement

In the assigned problem statement, the objective was to develop a Banking Information System capable of automating common banking activities and reducing dependency on manual processes.

Traditional banking management methods involve manual record keeping and handling of customer accounts and transactions, which may result in delays, data redundancy, operational inefficiencies, and increased chances of errors.

The project aimed to design and implement a centralized software solution capable of managing banking operations such as account creation, customer authentication, deposits, withdrawals, balance inquiry, fund transfer, account status management, transaction tracking, and reporting functionalities.

The system was developed using Core Java concepts and object-oriented programming techniques while implementing validations, collections, exception handling, and file handling mechanisms to simulate real-world banking operations.

4 Existing and Proposed solution

Traditional banking management systems often depend on manual record maintenance and paper-based operations for handling customer information and transactions. In smaller organizations and educational implementations, customer account details, transaction records, and balance information are often maintained manually.

Limitations of existing approaches:

• Time-consuming operations and record handling

• Increased possibility of human errors

• Difficulty in maintaining customer transaction history

• Lack of centralized account management

• Limited automation in banking operations

• Difficulty in generating reports and monitoring activities

• Poor scalability and maintenance issues

Proposed Solution

The proposed solution is a Banking Information System developed using Core Java to automate banking activities and improve operational efficiency.

The system provides a centralized platform to manage customer accounts, banking operations, authentication, and transaction management.

The developed system includes:

• Admin and customer authentication

• Account creation and management

• Deposit and withdrawal operations

• Balance inquiry functionality

• Fund transfer system

• Transaction history and mini statement generation

• Account status management

• Savings interest calculation
• Reporting and account sorting features
• Validation and exception handling mechanisms

Value Addition

The developed Banking Information System provides the following value additions:
• Automation of banking activities
• Reduced manual effort
• Improved data organization
• Better transaction tracking
• Validation-based secure operations
• Modular and scalable architecture
• Practical implementation of industrial software development concepts
• Better maintainability and usability

4.1 Code submission (Github link)

https://github.com/Avadhoot-007/upskillcampus/blob/main/BankingInformationSystem.java

4.2 Report submission (Github link) :

https://github.com/Avadhoot-007/upskillcampus/blob/main/BankingInformationSystem_Avadhoot_USC_UCT.pdf

5 Proposed Design/ Model

The Banking Information System was designed using a modular approach where different functionalities were divided into separate components to improve maintainability and scalability.

The system follows a structured workflow beginning from user authentication and proceeding toward banking operations and transaction processing.

The implementation was developed using Core Java concepts and object-oriented programming principles. Different modules such as authentication, account management, transaction handling, reporting, validations, and persistence mechanisms were integrated together to simulate real banking operations.

The system consists of the following major modules:

1. Authentication Module
2. Account Management Module
3. Transaction Management Module
4. Validation Module
5. Reporting Module
6. Persistence Layer

The overall process starts with authentication and then redirects users toward banking operations depending on their access role.

High Level Diagram (if applicable)

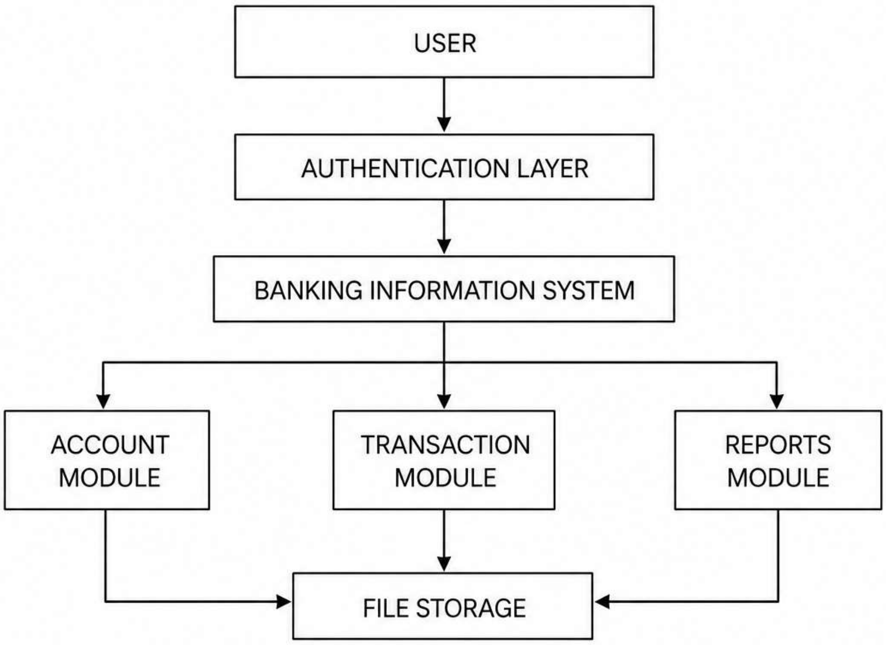


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.1 Low Level Diagram (if applicable)

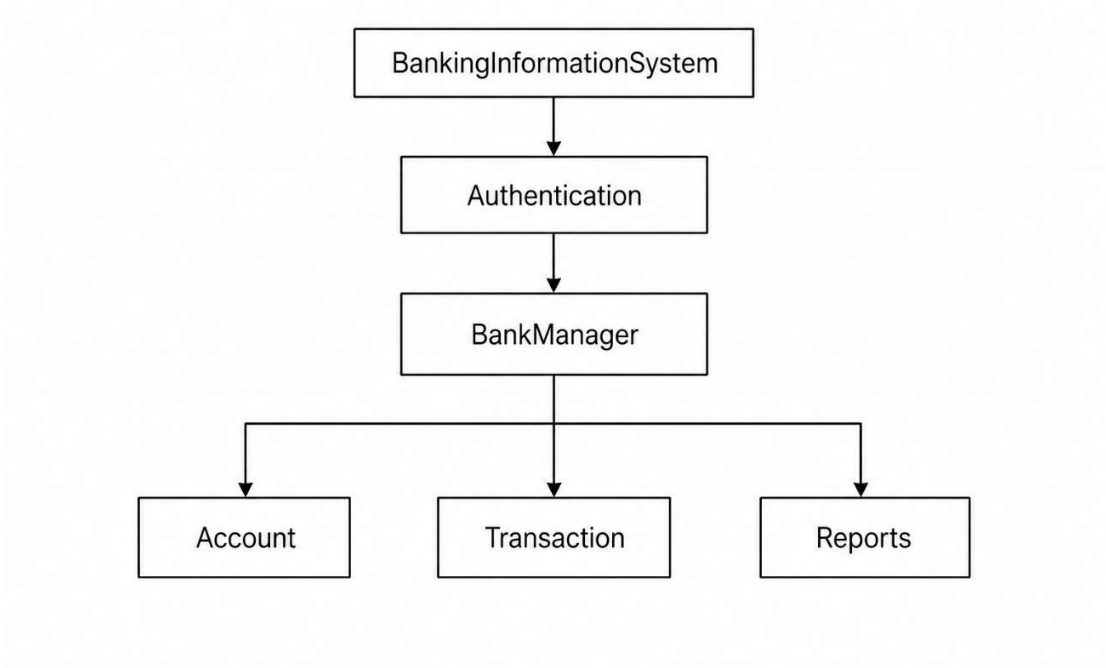


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.2 Interfaces (if applicable)

The Banking Information System uses a console-based interface to interact with users.

The system provides two major interfaces:

1. Admin Interface

Features:

• Account creation

• Account update

• Account deletion

• Search operations

• Reports generation

• Transaction monitoring

• Account sorting

• Status update

• Interest calculation

2. Customer Interface

Features:

• Deposit operations

• Withdraw operations

• Balance inquiry

• Fund transfer

• Mini statement generation

• Transaction history viewing

• Account information access

System Flow:

Main Menu

→ Authentication

→ User Selection

→ Banking Operations

→ Transaction Processing

→ Reports and Persistence

6 Performance Test

The performance testing phase was conducted to verify the reliability, correctness, and efficiency of the Banking Information System.

The developed system was tested against different operational constraints and user scenarios to ensure stable execution and proper validation handling.

The major constraints considered during testing were:

• Invalid user inputs

• Duplicate account entries

• Negative transaction values

• Insufficient account balance

• Invalid account access

• Inactive or blocked account handling

• Transaction processing validation

• Authentication validation

The system was tested to ensure that these constraints were handled properly through validations and exception handling mechanisms.

6.1 Test Plan/ Test Cases

Test Case 1

Test Objective:

Verify account creation functionality.

Input:

Valid account details.

Expected Result:

Account should be created successfully.

Outcome:

Passed.

Test Case 2

Test Objective:

Validate duplicate account prevention.

Input:

Existing account ID.

Expected Result:

System should reject duplicate account creation.

Outcome:

Passed.

Test Case 3

Test Objective:

Verify deposit functionality.

Input:

Valid deposit amount.

Expected Result:

Balance should increase.

Outcome:

Passed.

Test Case 4

Test Objective:

Verify withdrawal validation.

Input:

Amount greater than available balance.

Expected Result:

Transaction should be rejected.

Outcome:

Passed.

Test Case 5

Test Objective:

Verify transfer operation.

Input:

Valid source account, destination account and amount.

Expected Result:

Amount transferred successfully.

Outcome:

Passed.

Test Case 6

Test Objective:

Validate authentication mechanism.

Input:

Invalid credentials.

Expected Result:

Access denied.

Outcome:

Passed.

Test Case 7

Test Objective:

Verify account status validation.

Input:

Inactive / blocked account.

Expected Result:

Access restricted.

Outcome:

Passed.

6.2 Test Procedure

The testing process was performed in multiple stages.

Step 1:

Initialize the Banking Information System.

Step 2:

Create customer accounts using valid details.

Step 3:

Perform banking operations such as deposits, withdrawals, balance inquiry, and transfers.

Step 4:

Test validation scenarios using invalid inputs.

Step 5:

Verify authentication controls and account access.

Step 6:

Execute transaction operations and monitor updates.

Step 7:

Evaluate reporting and account management modules.

Step 8:

Verify persistence and interest handling mechanisms.

Step 9:

Record outputs and validate system behavior.

6.3 Performance Outcome

The Banking Information System successfully performed all major banking operations under testing conditions.

Observed outcomes:

- Successful account creation and management
- Correct transaction execution
- Proper balance updates
- Successful transfer operations
- Accurate validation handling
- Stable authentication behavior
- Controlled access for inactive accounts
- Proper exception handling
- Reliable execution of sorting and reporting operations
- Successful interest calculation for savings accounts

The system demonstrated stable execution and effectively handled operational constraints and user validations. The developed solution successfully fulfilled the project objectives and simulated practical banking workflows.

7 My learnings

The industrial internship provided valuable exposure to practical software development and industrial project implementation methodologies.

During the development of the Banking Information System, various technical and analytical concepts were learned and applied in practical scenarios.

Technical learnings gained during the internship include:

• Core Java programming concepts

• Object-Oriented Programming principles

• Collections framework usage

• File handling implementation

• Exception handling mechanisms

• Authentication and validation systems

• Transaction processing logic

• Modular software design

• Data organization and management

• Testing and debugging practices

The internship also helped in understanding project planning, requirement analysis, implementation workflows, and software documentation processes.

Apart from technical skills, the internship improved problem-solving ability, logical thinking, analytical skills, and confidence in handling real-world projects.

The exposure gained through this internship will contribute positively toward career growth and future software development opportunities.

8 Future work scope

The current Banking Information System successfully implements major banking operations and fulfills the project objectives. However, several enhancements can be implemented in future versions to improve functionality and scalability.

Possible future improvements include:

• Graphical User Interface (GUI) implementation

• Database integration using MySQL

• JDBC connectivity support

• Online banking functionality

• Mobile application integration

• Encryption and advanced security mechanisms

• Cloud deployment support

• Analytics and reporting dashboards

• Email and notification integration

• Multi-user concurrent operation handling

• Loan management module

• ATM simulation support

• Advanced transaction monitoring and analytics

Due to internship duration limitations, these enhancements were not implemented but can be considered for future development phases.